

Jingyi Le

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EDUCATION

University of Cambridge

MAST in Applied Mathematics (Part III)

Cambridge, UK

Oct 2025 – Jun 2026

- Interests: Number Theory (algebraic, arithmetic, analytic), Pure Dynamics; further in Algebra and Geometry

King's College London

Bachelor of Science(Hons), Mathematics

London, UK

Sep 2022 – May 2025

- GPA: 1st Class Honor (87); CGPA:89, 3.99/4.00
- Relevant Course:PDE, Number Theory, Real & Complex Analysis, Geometry of surfaces(1/114), Dynamics, Group, Rings & Modules, Fourier, Galois, Representation Theory, Probability Theory, Numerical

RESEARCH EXPERIENCE

p-adic Dynamics and Failure of Newton's Method

Research Intern (Supervisor: Prof. Holly Krieger)

University of Cambridge

Jul 2025 – Aug 2025

- Investigated Faber-Voloch's conjecture on Newton's failures, positive lower density and root bias
- Implemented large-scale computations in Mathematica to test p-adic dynamics and bad reductions
- Provided numerical evidence that apparent root bias vanishes at large prime bounds (still could exist)
- Proved existence of infinitely many primes of convergence and non-convergence in McMullen maps
- Extended research via McMullen and Shub-Smale maps to p-adic convergence, with superattracting fixed points
- Ongoing work; Manuscript in preparation (target: arXiv, Q4 2025)

Hubbard Tree and Entropy in Complex Dynamics

Research Intern (Supervisors: Dr. Harry Schmidt & Dr. Rob. Kropholler)

University of Warwick

Jul 2024 – Aug 2024

- Conducted a comprehensive review and proved key theorems on Riemann sphere, Julia sets and Fatou sets
- Modeled Julia Sets and Mandelbrot Sets with Mathematica, applied Newton's Method to quadratic polynomials
- Traced behaviors and structures of dynamics by post-critical finite Hubbard tree and its entropy algorithm
- Discussed progress and upcoming works with two assigned PhD students and the supervisors weekly
- Presented findings at Warwick CDT; outlined potential extensions

AWARDS & HONORS

Cambridge Open Research Fellowship | *awarded to 3 annually by CMS*

2025

Mary Lister McCammon Research Fellowship | *awarded by Imperial College London*

2025

The University of Hong Kong Summer Research Scholarship | *acceptance rate 5%*

2025

Warwick Math UROP Fellowship | *acceptance rate 10%*

2024

TALKS & PRESENTATIONS

'p-adic Dynamics and Failure of Newton's Method', CMP Day, Cambridge, UK

Aug 2025

WORKSHOP & CONFERENCE

Communicating Your Research: Writing Workshop, University of Cambridge

Aug 2025

BICMR Geometric Group Theory Summer School, Peking University

Jun 2025

SKILLS

Programming: Mathematica/ Python/ R

Typesetting: LaTeX (advanced, for research manuscripts)

Language: Mandarin (native)/ English (fluent)/ Cantonese (conversational)

Research notes, preprints, and diary available at: github.io